

# **Your Product Name**

A guide to vibration analysis and data management

This guide is based on Your Product v1.0.0.

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# System Requirements

## System Requirements

- ✓ Windows 10/11 or macOS 10.15+
- ✓ RAM: 8 GB minimum (16 GB recommended)
- ✓ Disk: 500 MB + data storage
- ✓ Browser: Chrome, Edge, Safari

## Quick Start

1. Run `start.sh` (Mac/Linux) or `start.bat` (Windows)
2. Open `http://localhost:8551` in your browser
3. Verify your data folder path (e.g., `D:/data/`)

# First Screen

## Screen Layout



Application screenshot

```
image("assets/main-screen.png")
```

Main application screen after launch

- ✓ **Navigation menu** — Switch between analysis modes
- ✓ **Data browser** — Browse files by machine, task, and date
- ✓ **Analysis area** — Charts and data visualization
- ✓ **Controls** — Adjust chart parameters and settings



Start with the Trend tab for your first analysis — it provides the best overview of your data patterns over time.

# Opening Your First File

## File Selection Steps

1. Select **Analysis > Single Channel** from the navigation menu
2. Choose the **Trend** tab
3. Select a machine from the **Machine** dropdown
4. Select a task from the **Work** dropdown
5. Date folders will load automatically
6. Select a date to see available time entries
7. Click a time entry to load channel data



Start with the Trend tab to see an overview of changes over time.



File browser screenshot

Data browser showing folder structure and file list

# Chart Analysis

## Reading a Trend Chart

- ✓ **X-axis:** Time (file creation timestamp)
- ✓ **Y-axis:** Vibration magnitude
- ✓ **Data points:** Peak value for each file

## Pattern Interpretation

- ✓ **Normal:** Consistent level maintained
- ✓ **Caution:** Sudden increase or irregular changes
- ✓ **Danger:** Sustained upward trend

## Analysis Tips



Focus on sudden changes rather than overall trends. Compare channels at the same time points for cross-reference analysis.



Large frequency ranges (e.g., 5–3000 Hz) increase processing time. Narrow the range when analyzing specific frequency bands.

# Cursor & Interaction

## Single Cursor

Click on the spectrum chart to place a cursor at a specific frequency.

- ✓ Exact frequency and amplitude display
- ✓ Fine-tune with arrow keys
- ✓ Track time history for that frequency

## Harmonics Cursor

Analyze integer multiples (1X, 2X, 3X...) of a base frequency.

- ✓ Automatic harmonic pattern overlay
- ✓ Detect imbalance and misalignment
- ✓ Bearing and gear fault diagnosis

## Sideband Cursor

Analyze sidebands at fixed intervals around a center frequency.

- ✓ Detect modulation patterns
- ✓ Configurable spacing interval
- ✓ Bearing and gear fault detection

# FAQ

## Q1: The chart is not displaying.

- ✓ Check frequency range settings
- ✓ Verify channel selection
- ✓ Confirm data file exists

## Q3: New folder/files not visible.

- ✓ Wait 5 minutes and retry (cache refresh)
- ✓ Refresh the browser
- ✓ Check folder structure
- ✓ Restart the system

## Q2: Loading takes too long.

- ✓ Check file count (single: 1000, multi: 300)
- ✓ Narrow frequency range (wider = slower)
- ✓ Refresh the browser

## Q4: Memory error appears.

- ✓ Reduce number of channels (2-3 at a time)
- ✓ Narrow file range
- ✓ Restart the browser



# Thank you.

## Technical Support

- Email: [support@example.com](mailto:support@example.com)
- Phone: +1-555-0100